Artificial Intelligence COMP9414 26T2

M11A, H18A tutorial

13/07/2026 Monday

Content of today:

1. How do you feel about assignment1 discussion
2. Markov Decision Processes (MDP)
3. TD prediction and difference between Q-learning and SARSA
4. Project show

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How do you feel about assignment1 discussion

Supervised Learning

Supervised Learning algorithms tried to make their outputs mimic the labels y given in the training set.

Supervised Learning
(e.g. classification task)

Prior knowledge of classes

dimoo | twinkle twinkle | labubu | crybaby | zsiga | molly

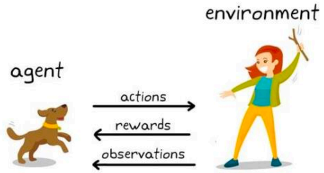
Supervised learning

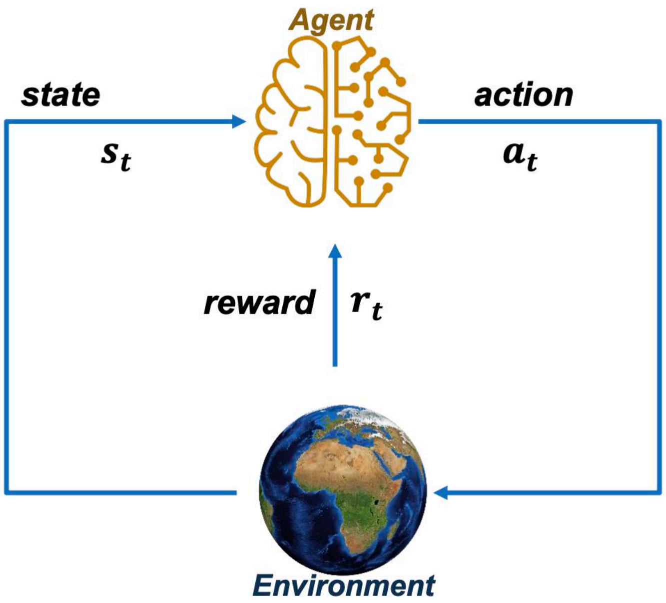
→ Predictive model → labubu

Each input x was paired with a label indicating the correct output.

Reinforcement Learning

What is a reinforcement learning?





At each step t the agent:

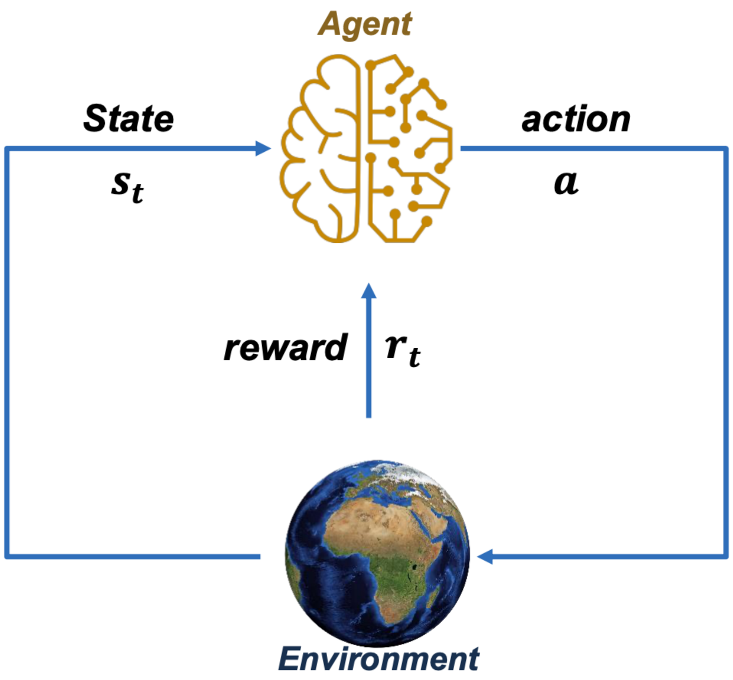
- *Executes action* $\boldsymbol{a_t}$
- *Receives state* $\boldsymbol{s_t}$
- *Receives scalar reward* $\boldsymbol{r_t}$

The environment:

- *Receives action* $\boldsymbol{a_t}$
- *Emits state* $\boldsymbol{s_{t+1}}$
- *Emits scalar reward* $\boldsymbol{r_{t+1}}$

t increments at env. step

Markov Decision Process



$$(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathbb{P}, \gamma)$$

$\mathcal{S}$: set of possible states
$\mathcal{A}$: set of possible actions
$\mathcal{R}$: distribution of reward given (state, action) pair
$\mathbb{P}$: **transition probability**
i.e. distribution over next state given (state, action) pair

γ : discount factor

History

$$H_t = s_1, r_1, a_1, \ldots, a_{t-1}, s_t, r_t$$

Markov Decision Process

Major Components of an RL Agent

An RL agent may include one or more of these components:

› Policy: agent’s behaviour function
› Value Function: how good is each state and/or action
› Model: agent’s representation of the environment

Markov Decision Process

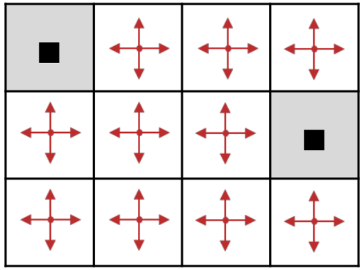
Let's try to work on a simple Markov Decision Process (MDP) example

"Reach one of terminal states (greyed out) in least number of actions"

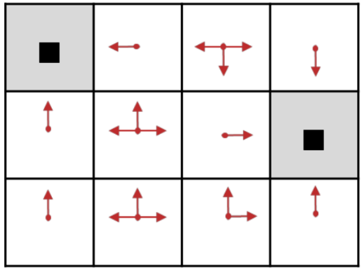
actions = {
- right →
- left ←
- up ↑
- down ↓

}

Random Policy



Optimal Policy



****Policy:*** *agent's behaviour function*

Markov Decision Process

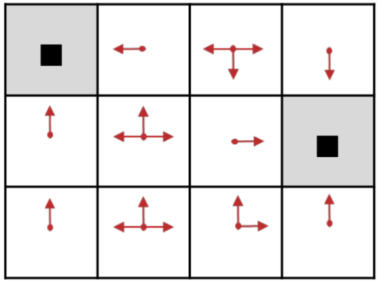
Definition: Policy and the optimal policy π*

› *We want to find optimal policy π* that maximizes the sum of rewards.*

› *How do we handle the randomness (initial state, transition probability…)?*

- ***Maximize the expected sum of rewards!***

Optimal Policy



Formally: $\pi^* = \arg\max_{\pi} \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t | \pi\right]$ with $s_0 \sim p(s_0), a_t \sim \pi(\cdot|s_t), s_{t+1} \sim p(\cdot|s_t, a_t)$

Markov Decision Process

Definitions: Value function and Q-value function

› *Following a policy produces sample trajectories (or paths) $s_0, a_0, r_0, s_1, a_1, r_1, \ldots$*

› ***State-Value Function*** *(How good is a state?)*

- *The value function at state s, is the expected cumulative reward from following the policy from state s:*

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t | s_0 = s, \pi\right]$$

› ***Action-Value Function*** *(How good is a state-action pair?)*

- *The Q-value function at state s and action a, is the expected cumulative reward from taking action a in state s and then following the policy:*

$$Q^{\pi}(s,a) = \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t | s_0 = s, a_0 = a, \pi\right]$$

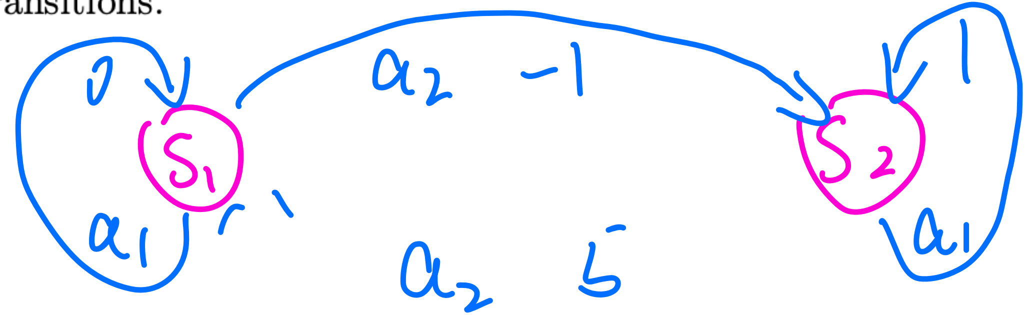
RL exercise

Question 1: Value functions

Consider a world with two states $S = \{S_1, S_2\}$ and two actions $A = \{a_1, a_2\}$, where the transitions δ and reward r for each state and action are as follows:

$$
\begin{array}{ll}
\delta(S_1, a_1) = S_1 & r(S_1, a_1) = 0 \\
\delta(S_1, a_2) = S_2 & r(S_1, a_2) = -1 \\
\delta(S_2, a_1) = S_2 & r(S_2, a_1) = +1 \\
\delta(S_2, a_2) = S_1 & r(S_2, a_2) = +5
\end{array}
$$

i. Draw a picture of this world, using circles for the states and arrows forthe transitions.



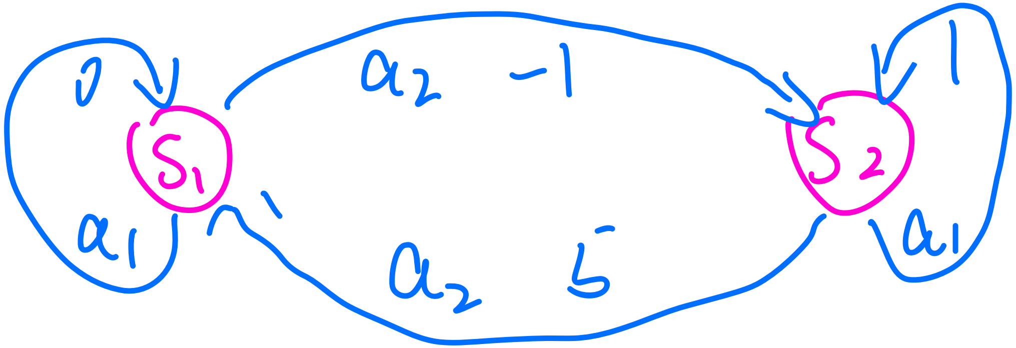
RL exercise

ii. Assuming a discount factor of $\gamma = 0.9$, determine:

(a) the optimal policy $\pi^* : S \to A$

(b) the state-value function $V^* : S \to R$

(c) the action-value function $Q^* : S \times A \to R$



(a) $V^{a_1}(s_1) = 0 + \gamma \cdot 0 + \gamma^2 0 + \gamma^3 0 \cdots = 0$

$\star$ $V^{a_2}(s_1) = -1 + \gamma(5 + \gamma(-1 + \gamma(5 \cdots$

$= -1 + \gamma \cdot 5 + \gamma^2 \cdot -1 + \gamma^3 5 \cdots$

$V^{a_1}(s_2) = \underbrace{1 + \gamma 1}_{1.9} + \gamma^2 1 + \gamma^3 1 \cdots$

$\star$ $V^{a_2}(s_2) = \underbrace{5 + \gamma(-1}_{4.1} + \gamma(5 + \gamma(-1 \cdots$

(b) the state value function $V^*: S \to R$

$$V^*(s) = R(s, a) + \gamma \cdot V^*(s')$$

$$\begin{cases} V^*(s_1) = -1 + \gamma \cdot V^*(s_2) & \text{①} \\ V^*(s_2) = 5 + \gamma \cdot V^*(s_1) & \text{②} \end{cases}$$

$$V^*(s_1) = -1 + \gamma \cdot (5 + \gamma \cdot V^*(s_1))$$

$$= -1 + 5\gamma + \gamma^2 \cdot V^*(s_1)$$

$$= \frac{-1 + 5\gamma}{\quad} = \frac{-1 + 4.5}{\quad} = 18.45$$

$(1-\gamma^2)$ $\quad 0.19$

$$V^*(S_2) = 5 + 0.9 \cdot 18.95 = 22.05$$

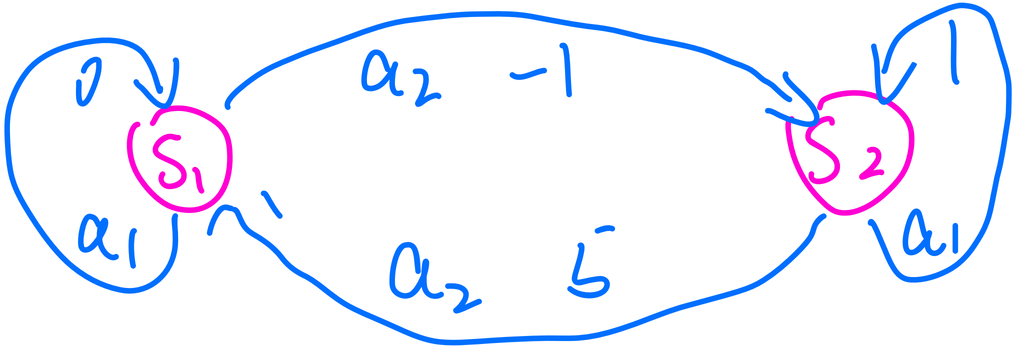
(c) $Q(s, a) = R(s, a) + \gamma \cdot V^*(s')$

$Q(s_1, a_1) = 0 + 0.9 \cdot V^*(s_1) =$

$Q(s_1, a_2) = 18.95$

$Q(s_2, a_1) = 1 + 0.9 \cdot V^*(s_2) =$

$Q(s_2, a_2) = 22.05$



S_1
S_2
a_1
0
a_2
-1
a_2
5
a_1
1

RL exercise

iii. Write the Q-values in a table (a.k.a. Q-table) as follows:

Q	a_1	a_2
S_1		
S_2		

RL exercise

iv. Trace through the first few steps of the action-value function learning algorithm, with all Q-values initially set to zero. Explain why it is necessary to force exploration through probabilistic choice of actions in order to ensure convergence to the true Q-values.

Reinforcement Learning

$V = r + \gamma(V')$

Consider a reinforcement learning environment with three states S= $\{S_1, S_2, S_3\}$ and two actions A= $\{a_1, a_2\}$. The transition probabilities and rewards are as follows:

State	Action	Next State	Reward
S_1	a_1	S_2 (70%), S_3 (30%)	2
S_1	a_2	S_3 (100%)	1
S_2	a_1	S_1 (100%)	0
S_2	a_2	S_2 (80%), S_3 (20%)	3
S_3	a_1	S_1 (50%), S_2 (50%)	-8
S_3	a_2	Terminal (100%)	4

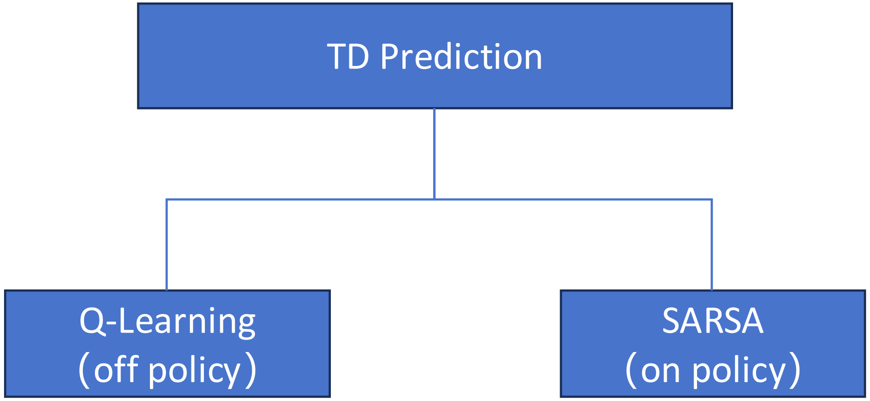
Assuming a discount factor γ = 0.9, what is the expected value of taking action a_1 in state S_1, given that the optimal policy will be followed thereafter? (Round your answer to two decimal places)

A. 15.88 B. 14.07 C. 13.29 D. 12.34 E. 11.45

Temporal-Difference (TD) Prediction

TD Prediction is a core method in reinforcement learning for **estimating the value of a state** — meaning how much total reward the agent can expect to collect in the future starting from that state.

It **learns from partial experience**: updating estimates based on what actually happens **after one step**, instead of waiting for a full episode to finish.



Difference between Q-learning and SARSA

State:(1,0) next action: →

SARSA

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\right]$$

Q[(1,0), →]

↑

Q[(1,0), →] + α × (reward + γ × Q[(1,1), next_action] - Q[(1,0), →])

Where:

next_action selected by ε-greedy

Q-Learning

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\left[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)\right]$$

Q[(1,0), →]

↑

Q[(1,0), →] + α × (reward + γ × max(Q[(1,1), :]) - Q[(1,0), →])

Where:

Q[(1,1), ↑] = -0.2 # up: normal way

Q[(1,1), ↓] = -0.5 # down： drop

Q[(1,1), ←] = 0.1 # left： back to (1,0)

Q[(1,1), →] = 0.3 # right： heading to G

So max(Q[(1,1), :]) = 0.3

RL exercise

Question 2: Temporal-difference learning

Consider the same world as the previous question. Assume the use of temporal-difference learning with the following parameters: learning rate $\alpha = 0.3$, discount factor $\gamma = 0.9$, and ϵ-greedy action selection method with $\epsilon = 0.1$. After a few steps of iterating, the learning agent performs action a_1 from state S_1 with the Q-table containing the following values:

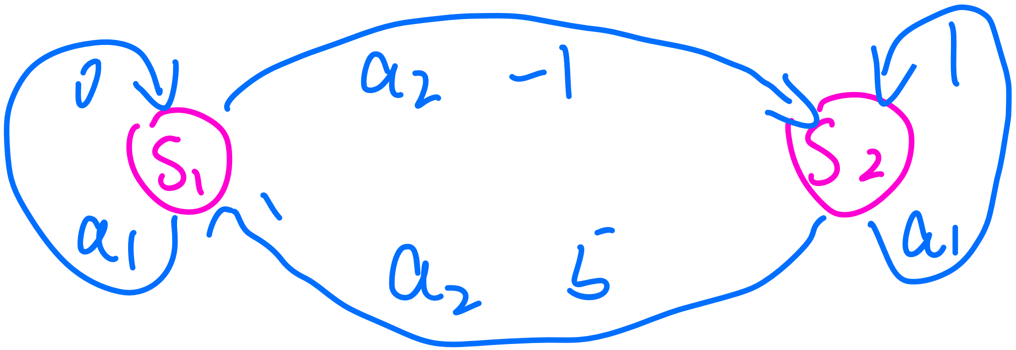
Q	a_1	a_2
S_1	0.15	3.55
S_2	5.72	9.18

i. How would look the Q-table after one iteration of the off-policy method Q-learning?

ii. How would look the Q-table after one iteration of the on-policy method Sarsa? Assume a random value $rnd = 0.01$.

iii. Explain how differently would the on-policy Sarsa method converge to the optimal value function in comparison to off-policy Q-learning.

```
if random() < epsilon:
    # Exploration — choose a random action
else:
    # Exploitation — choose the best known
                    action (max Q value)
```



① Q-learning

$$Q(S_t, a_t) \leftarrow Q(S_t, a_t) + \alpha\left[R_t + \gamma \max_{action(A)} Q(S_{t+1}, a) - Q(S_t, a_t)\right]$$

$$Q(S_1, a_1) \leftarrow Q(S_1, a_1) + \alpha\left[0 + \gamma \cdot 3.55 - Q(S_1, a_1)\right]$$

② SarSa

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_t + \gamma\, Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t) \right]$$

$$\begin{cases} a_1 = 0.15 \\ a_2 = 3.55 \end{cases}$$

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~Questions and Suggestions~

Thanks for listening

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